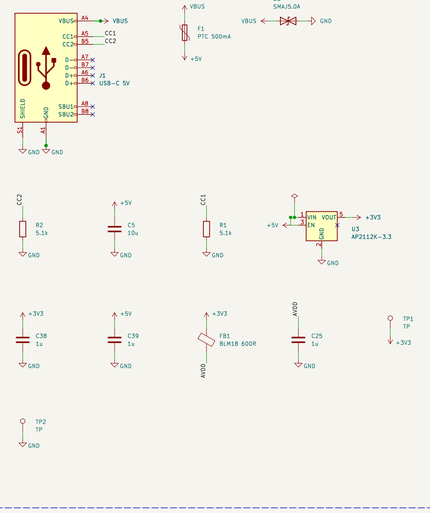
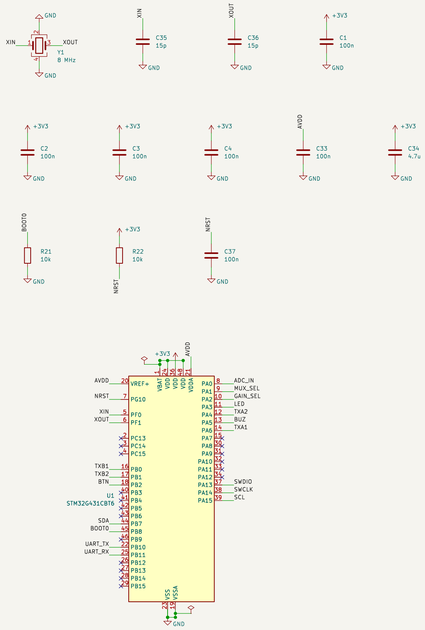
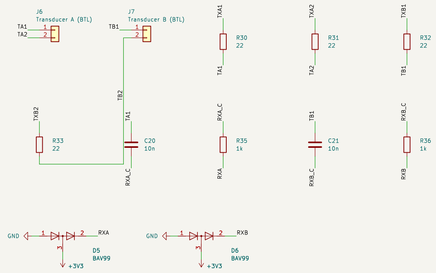
USB-C 5 V INPUT + 3.3 V LDO + filtered analog supply



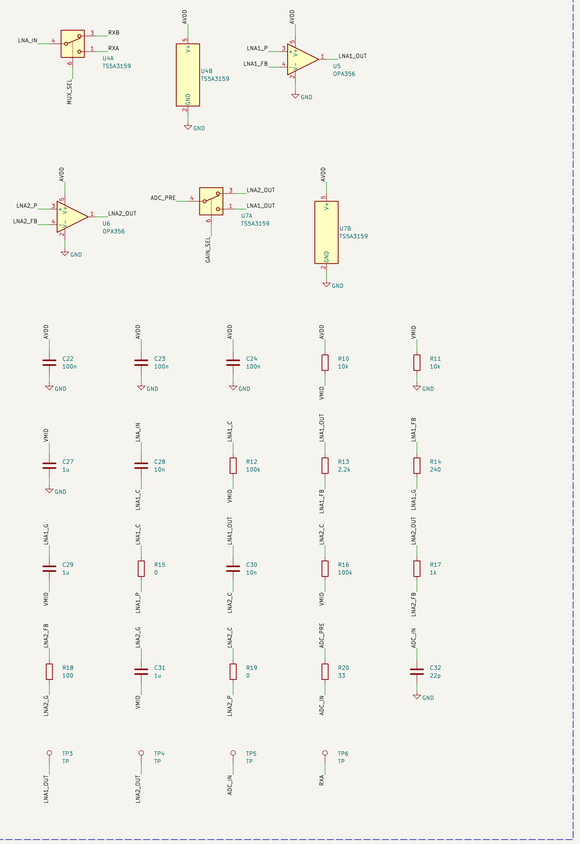
MCU: STM32G431CBT6 (170 MHz, 12-bit ADC 4 MSPS, 8 MHz crystal)



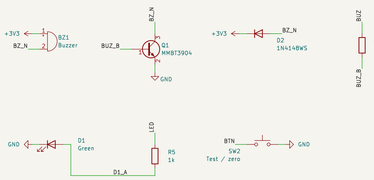
TRANSDUCER DRIVE: 22 ohm series, BTL from two timer pins, clamp + AC-coupled receive tap



RECEIVE CHAIN: input mux, 2 x OPA356 (x10 each), gain-select mux, ADC anti-alias



ALARM, LED, BUTTON



OLED, UART, SWD



Measurement: STM32 TIM3 makes a 6-cycle 1 MHz burst on two pins (second inverted = bridge drive, 6.6 Vpp) into transducer A; transducer B receives through U4/LNA/U7 into ADC1 at 4 MS/s (64 samples DMA). Then roles swap. 1000 up/down pairs per measurement.

While receiving, one leg of the transducer is held low by its timer pin and the other is high-impedance and feeds the clamp + mux. LNA1 x10.2 (20 dB), LNA2 x10 (+20 dB), U7 selects 20 or 40 dB.

2-layer prototype. Production should use 4 layers (solid analog ground, guarded receive trace) — see docs.

First engineering release; see pcb/README.md for status and limits

ClaudeWork

Sheets: /

File: jal_netra.kicad_sch

Title: Jal-Netra ultrasonic flow main board

Size: A1	Date: 2026-11-01	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1